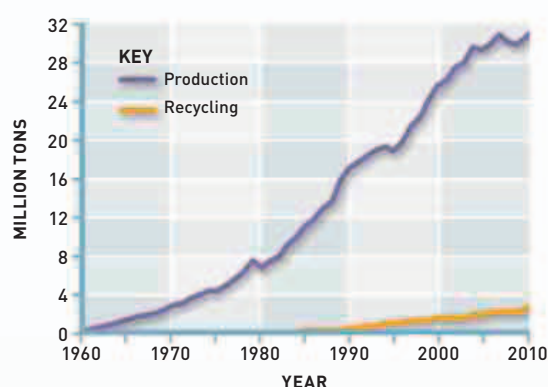


THE STORY OF PLASTICS

BY THE END OF THE 20TH CENTURY THE AGE OF PLASTICS HAD ARRIVED, TRANSFORMING MANY INDUSTRIES AND THE HOME

Plastic is one of the most remarkable of all man-made materials, used in everything from spaceships and computers to bottles and artificial body parts. What gives plastic its special quality is the shape of its molecules. Most plastics are made from long organic molecules known as polymers.

In the mid-19th century, people knew that cellulose (the woody substance in plants) could be made into a brittle substance called cellulose nitrate. In 1862, British chemist Alexander Parkes added camphor to it, producing a tough but moldable plastic called parkesine. In 1869, American inventor John Hyatt created a similar substance called celluloid, which was used to make photographic film by Kodak in 1889. Today there are thousands of synthetic plastics, each with their own properties and uses. Many are still based on hydrocarbons (oil or natural gas), but in recent decades carbon fibers and other materials have been added to create superlight, superstrong plastics such as Kevlar and CNRP.



Production and recycling of plastics

In recent years, there have been concerted attempts to recycle plastics, made easier by the establishment of recycling centers and collection services. However, the slight rise in recycled plastic lags far behind the soaring production of new plastics, as this graph clearly shows.



RECYCLING

Plastics are widely used because they are durable and tough, but they are not biodegradable. Once disposed of, they linger in the environment for a very long time. The vast amount of waste plastic now in the oceans—maybe hundreds of millions of tons—is damaging marine wildlife. It is important to reduce plastic use and recycle as much as possible. Not all plastics can be recycled easily, and it takes energy to heat the plastic to reform it, so recycling rates are low.

“ I THOUGHT I SHOULD MAKE SOMETHING REALLY SOFT INSTEAD THAT COULD BE MOLDED INTO DIFFERENT SHAPES. ”

Leo Baekeland, Belgian chemist, on inventing bakelite

1926

Vinyl

American chemist Waldo Semon exposes PVC to heat and a range of chemicals to produce vinyl. It is used to make objects ranging from shoes to shampoo bottles.

1935

Nylon

American chemist Wallace Carothers invents nylon, the first thermoplastic—it is liquid when hot and sets hard when it cools. Best known for its use in stockings, it has many other uses.

Nylon toothbrush



1937

Teflon

PTFE, or Teflon, is invented by American chemist Roy Plunket. It is not made from hydrocarbons but from fluorine joined to carbon, and is often used in frying pans.



Teflon frying pan

1966

Kevlar

American chemist Stephanie Kwolek spins heat-resistant fibers from liquid hydrocarbons. These fibers can be woven together to make materials such as Kevlar.



Bulletproof Kevlar

1933

Polyethylene

British chemists Eric Fawcett and Reginald Gibson create practical polyethylene in 1933, although it was first made in 1898. It is tough, soft, and flexible, and now the most widely used of all plastics.

Polyethylene crop tunnel



1936

Polystyrene

Styrol is the oily substance in the resin of Turkish sweetgum trees. In 1936 German chemical company I G Farben use it to produce polystyrene.

1954

Polypropylene

This rugged plastic resists many solvents and acids. It has uses ranging from wrapping to bottles for medical chemicals.

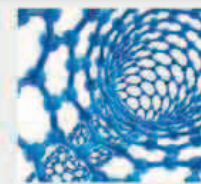


Polypropylene rope

1991

CNRP

Japanese physicist Sumio Iijima rolls carbon molecules into nanotubes. These can reinforce plastic to make strong, light CNRP.



Carbon nanotube molecule